PRINTABLE QUICK GUIDE

Roll your own seed phrase.

A paper-first summary for converting physical dice rolls into a 12- or 24-word BIP-39 seed phrase.

Use this printed guide for a real seed phrase. If you are unsure about the method, first use the online guide or make a dry run with a dummy phrase. Keep rolls, bits, and seed words off networked devices.

Prepare

Dice

Plain, well-made six-sided dice. Avoid decorative, novelty, damaged, or visibly uneven dice.

Rolling setup

A lidded box with enough room for every die to move freely, or a large cup and hard rolling tray.

Printouts

One Bit-to-word worksheet and the searchable BIP-39 word list.

Pen

Write every converted bit on paper. Never type rolls into a connected device.

Offline checksum device

A hardware wallet or permanently offline laptop that supports checksum completion.

Private room

No cameras, phones, smart glasses, observers, or recording devices.

1

Choose your phrase length

Choose before rolling. Base-4 works with 12 or 24 words. With regular dice, use 24 words for binary quantization.

Print the matching worksheet and write a **non-secret identifier** on it so you can recognize the backup later.

2

Choose one encoding

Base-4 rejection sampling — recommended

- 1 → `00`
- 2 → `01`
- 3 → `10`
- 4 → `11`
- 5 or 6 → skip

Collect exactly 64 accepted results for 12 words, or 128 for 24 words.

Binary quantization

- 1, 2, or 3 → `0`
- 4, 5, or 6 → `1`

For 12 words, use casino-grade dice. With regular dice, choose 24 words or use base-4. Make exactly 128 rolls for 12 words, or 256 for 24 words.

3

Roll and record

Lidded box: Put all dice in the box, close the lid, and shake vigorously at least ten times. Each shake should make the dice travel and collide audibly inside. Put the box down, gently tip the settled dice to one edge, then read left to right (and front to back for another row). Do not arrange dice by hand.

Cup and tray: Shake vigorously, cast with enough room and energy for several impacts, and read dice by a spatial rule chosen before the first throw.

Convert every result with the encoding you chose and write the bits into the next empty boxes. If any die is cocked or ambiguous, discard the **whole batch** and roll that complete batch again.

Stop at the bold divider in the final word. Leave the final 4 boxes blank for a 12-word phrase, or the final 8 boxes blank for a 24-word phrase. In that last batch, retain results only until the boundary—never select results by value.

4

Convert bits to words

For each completed 11-bit group, use the printed BIP-39 word list:

- bits 1–3 select the **page**;
- bits 4–6 select the **column**;
- bits 7–11 select the **row** and therefore the word.

Write the word beneath its 11 bits. Do not search the word list digitally, even for a partial bit pattern.

5

Complete the final word

The final word has checksum bits that are determined by the bits already recorded. They do not reduce the randomness supplied by the dice.

- For 12 words, your seven rolled final-word bits leave **16** possible words.
- For 24 words, your three rolled final-word bits leave **256** possible words.

Depending on the wallet, enter the earlier words as an existing seed phrase. Let it propose a final word, then verify that the proposed word remains in your allowed range.

Keep your seed phrase private, offline, and backed up safely.